

EXP NO:7A	DATA VISUALIZATION USING TABLEAU
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Aim:

To import, transform, analyze and visualize the Baseball Databank using Tableau Desktop and create interactive worksheets and dashboards for team, season and home-run analysis.

Algorithm:

- 1 Connect the Baseball batting dataset to Tableau.
- 2 Verify fields including playerID, yearID, teamID, total_runs and HR.
- 3 Create a calculated field for Total Runs.
- 4 Create a bar chart for the top 10 teams by total runs.
- 5 Create a line chart for total runs by year.
- 6 Create a visualization for home runs by team.
- 7 Create a Minimum Runs parameter and apply it to team filtering.
- 8 Combine the visualizations into an interactive dashboard.

Output:

```
In [1]: from pathlib import Path

def find_data(filename):
    candidates = [
        Path(filename),
        Path("data") / filename,
        Path("../data") / filename,
        Path("../..data") / filename,
        Path("/content") / filename,
    ]
    for p in candidates:
        if p.exists():
            return p
    raise FileNotFoundError(f"{filename} not found. Upload the Baseball Databank CSV or keep it in the repository data folde

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv(find_data("Batting.csv"))
df["total_runs"] = df["R"]
df["is_home_run"] = (df["HR"] > 0).astype(int)

print("Baseball dataset loaded successfully")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
display(df.head())
```

Baseball dataset loaded successfully
Rows: 101332
Columns: 24

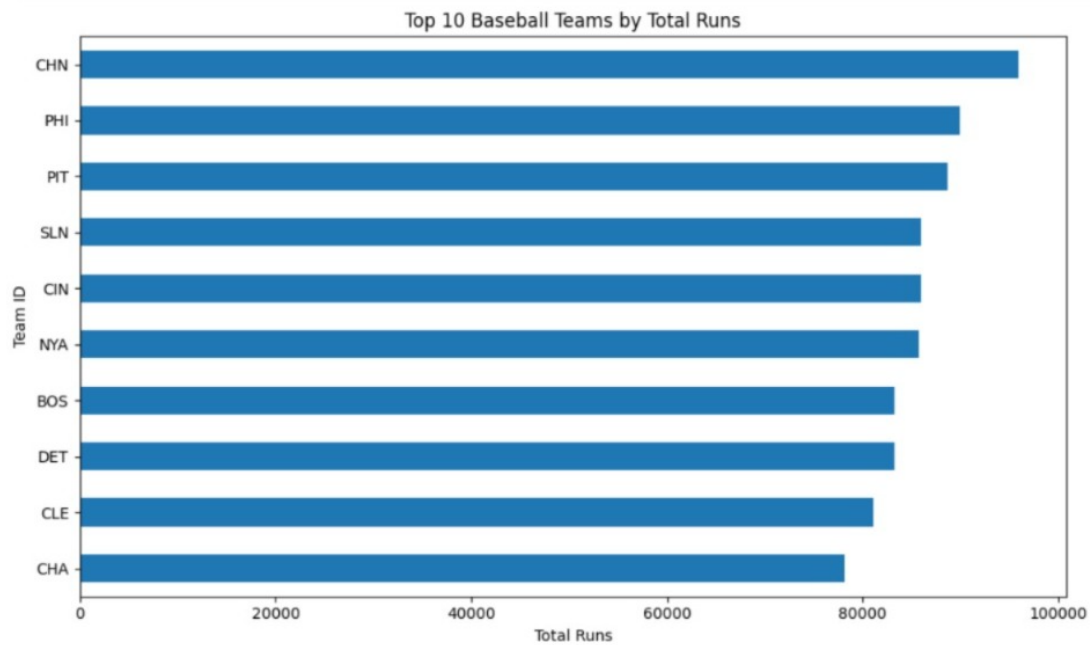
	playerID	yearID	stint	teamID	lgID	G	AB	R	H	2B	3B	HR	RBI	SB	CS	BB	SO	IBB	HBP	SH	SF	GIDP	tot
0	abercda01	1871	1	TRO	NaN	1	4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	NaN	NaN	NaN	NaN	NaN	
1	addybo01	1871	1	RC1	NaN	25	118.0	30.0	32.0	6.0	0.0	0.0	13.0	8.0	1.0	4.0	0.0	NaN	NaN	NaN	NaN	NaN	
2	allisar01	1871	1	CL1	NaN	29	137.0	28.0	40.0	4.0	5.0	0.0	19.0	3.0	1.0	2.0	5.0	NaN	NaN	NaN	NaN	NaN	
3	alliso01	1871	1	WS3	NaN	27	133.0	28.0	44.0	10.0	2.0	2.0	27.0	1.0	1.0	0.0	2.0	NaN	NaN	NaN	NaN	NaN	
4	ansonca01	1871	1	RC1	NaN	25	120.0	29.0	39.0	11.0	3.0	0.0	16.0	6.0	2.0	2.0	1.0	NaN	NaN	NaN	NaN	NaN	

```
In [2]: team_runs = df.groupby("teamID")["total_runs"].sum().sort_values(ascending=False)
year_runs = df.groupby("yearID")["total_runs"].sum()

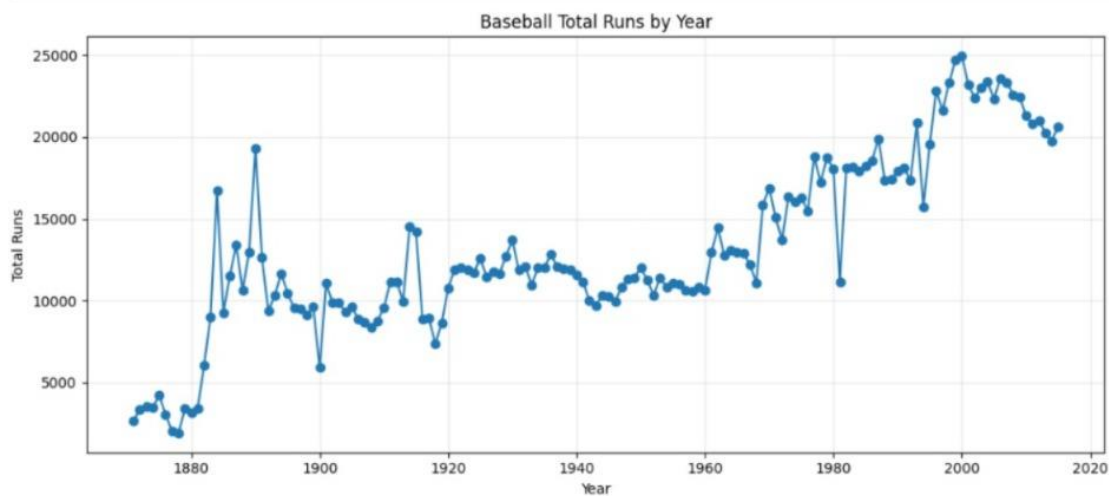
print("Calculated fields:")
print("Total Runs = SUM([total_runs])")
print("Home Runs = SUM([HR])")
print("Minimum Runs parameter can be applied to filter teams.")
```

Calculated fields:
Total Runs = SUM([total_runs])
Home Runs = SUM([HR])
Minimum Runs parameter can be applied to filter teams.

```
In [3]: plt.figure(figsize=(10,6))
team_runs.head(10).sort_values().plot(kind="barh")
plt.title("Top 10 Baseball Teams by Total Runs")
plt.xlabel("Total Runs")
plt.ylabel("Team ID")
plt.tight_layout()
plt.savefig("Experiment_7A_Baseball_Team_Bar.png", dpi=150, bbox_inches="tight")
plt.show()
```

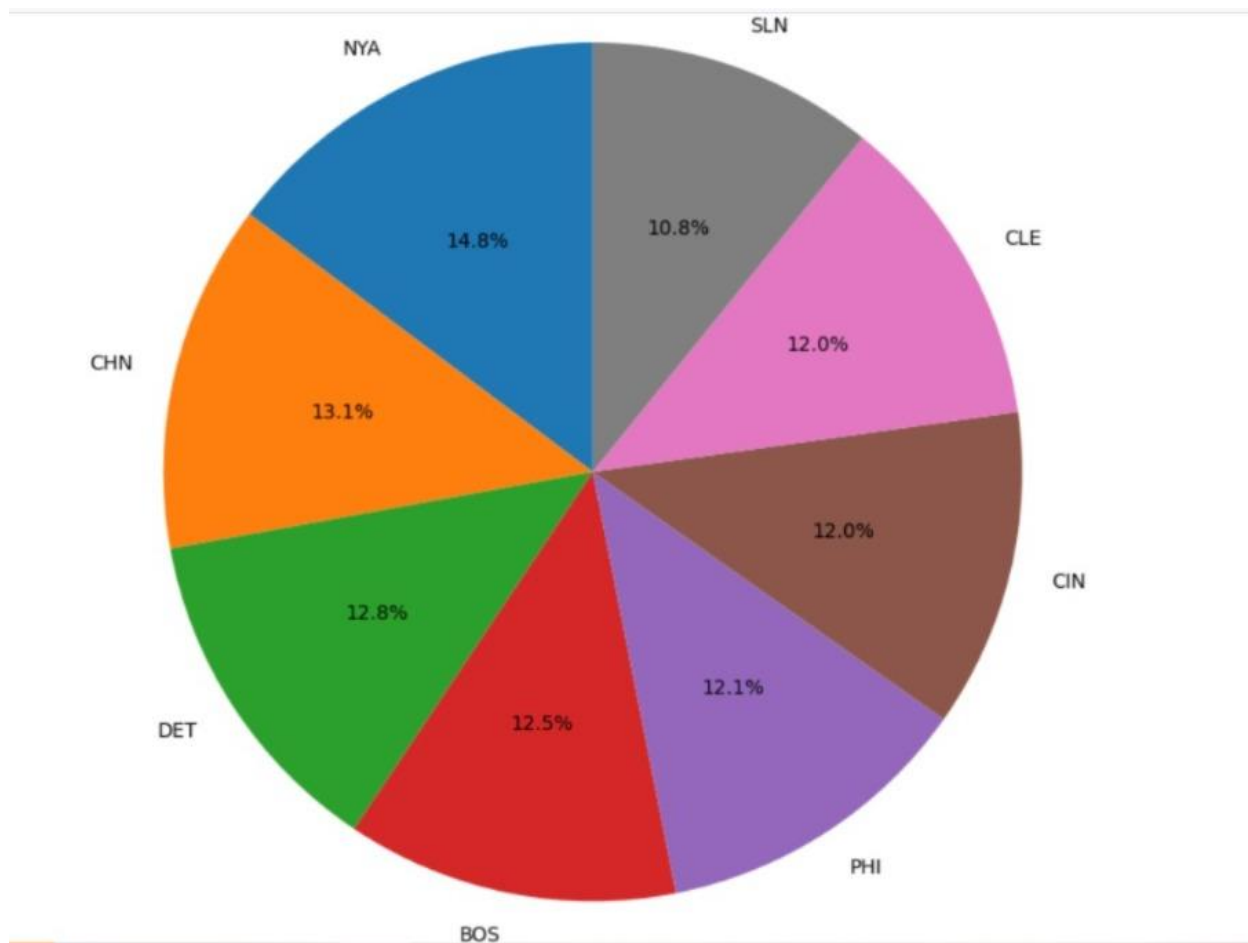


```
In [4]: plt.figure(figsize=(11,5))
year_runs.plot(kind="line", marker="o")
plt.title("Baseball Total Runs by Year")
plt.xlabel("Year")
plt.ylabel("Total Runs")
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig("Experiment_7A_Baseball_Year_Line.png", dpi=150, bbox_inches="tight")
plt.show()
```



```
In [5]: home_runs_by_team = df.groupby("teamID")["HR"].sum().nlargest(8)
plt.figure(figsize=(8,8))
plt.pie(home_runs_by_team.values, labels=home_runs_by_team.index, autopct="%1.1f%%", startangle=90)
plt.title("Home Run Distribution Across Top Teams")
plt.tight_layout()
plt.savefig("Experiment_7A_Baseball_HomeRun_Pie.png", dpi=150, bbox_inches="tight")
plt.show()
```

Home Run Distribution Across Top Teams



```
In [6]: minimum_runs = 10000
filtered_teams = team_runs[team_runs >= minimum_runs].sort_values(ascending=False)

print(f"Minimum Runs parameter = {minimum_runs:,}")
print("Teams meeting the parameter:")
display(filtered_teams.to_frame("Total Runs"))
```

Minimum Runs parameter = 10,000
Teams meeting the parameter:

Total Runs	
teamID	
CHN	96003.0
PHI	89959.0
PIT	88658.0
SLN	86003.0
CIN	85970.0
NYA	85715.0
BOS	83309.0
DET	83222.0
CLE	81116.0
CHA	78221.0
NY1	54282.0
BSN	48757.0
BRO	47236.0

PHA	36497.0
HOU	35783.0
NYN	35074.0
OAK	34616.0
ATL	34593.0
SLA	33546.0
KCA	32982.0
TEX	32927.0
SDN	29986.0
TOR	28712.0
SEA	27490.0
MON	23350.0
CAL	20766.0
ML4	19564.0
COL	18512.0
WAS	13682.0
FLO	13300.0
ARI	13153.0
MIL	12961.0
TBA	12859.0
LAA	10882.0

Result:

Thus, the Baseball Databank was successfully connected and analyzed in Tableau using team, season and home-run visualizations and an interactive Minimum Runs parameter.